

PROPOSAL IDE/CONCEPT PAPER SAMSUNG SOLVE FOR TOMORROW 2026

Tulis ide solusi yang ingin Anda ajukan dengan mengisi kolom-kolom di bawah secara lengkap dan spesifik! Jika ada pertanyaan, silakan hubungi di WhatsApp Enrico (6289677878563) atau di email: sft2026@dibimbing.id atau dapat bergabung ke community Telegram Samsung Solve for Tomorrow 2026.

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Nama Anggota 1	: <i>(Anggota Tim 1)</i>
Nama Anggota 2	: <i>(Anggota Tim 2)</i>

1. JUDUL PROJEK/IDE

Water Presence Monitoring System: Citizen Science + AI-Powered Remote Sensing untuk Deteksi dan Monitoring Real-Time Keberadaan Air Permukaan

2. TEMA/BIDANG YANG DIPILIH

Sustainability & Environment

3. LATAR BELAKANG

Konteks Masalah (Data & Fakta)

Indonesia sebagai negara dengan geografis kepulauan yang luas menghadapi tantangan serius dalam monitoring dan deteksi sumber daya air. Berdasarkan data dari Kementerian Pekerjaan Umum dan Perumahan Rakyat (PUPR), akses ke data water resources yang akurat dan real-time masih terbatas, terutama di daerah terpencil dan pasca-bencana.

Data menunjukkan:

- Lebih dari 40% daerah Indonesia masih kesulitan dalam pemetaan sumber daya air permukaan
- Survei manual untuk mendeteksi keberadaan air di area bencana memakan waktu 3-5 hari
- Satelit konvensional (Landsat) memiliki resolusi 30m, tidak cukup untuk area lokal
- Biaya survey ground-based berkisar 5-10 juta rupiah per lokasi
- Tidak ada platform yang mengintegrasikan citizen observations dengan data satelit secara real-time

- Citra satelit sering terhalang cloud cover (30-50% tidak dapat digunakan), menyebabkan keterlambatan data

Riset Pengguna (Persona & Kutipan Wawancara)

Persona 1: Disaster Response Manager

- Profil: Koordinator bencana di Badan Nasional Penanggulangan Bencana (BNPB)
- Tantangan: “Kami butuh informasi cepat tentang area yang tergenang untuk prioritas evakuasi, tapi data satelit lambat dan mahal”
- Kebutuhan: Pemetaan real-time area tergenang dalam hitungan menit, bukan hari

Persona 2: Water Resource Manager

- Profil: Petugas dinas pertanian di kabupaten rural
- Tantangan: “Susah monitor ketersediaan air di area pertanian kami yang luas tanpa data akurat”
- Kebutuhan: Indikator ketersediaan air permukaan per musim untuk perencanaan irigasi

Persona 3: Environmental Scientist

- Profil: Peneliti di universitas yang fokus pada environmental monitoring
- Tantangan: “Data satellite mahal dan jarang tersedia cloud-free, perlu alternative untuk baseline assessment”
- Kebutuhan: Tool yang accessible dan dapat di-crowdsource untuk citizen science

Pernyataan Masalah Inti (How Might We...)

“How might we create an accessible, real-time water presence detection system that combines smartphone observations with satellite validation and AI analysis, enabling rapid assessment for disaster response, agricultural planning, and environmental monitoring?”

4. TARGET PENGGUNA

Primary Users

Petugas disaster response (BNPB),
Water resource managers (dinas
pertanian/PUPR), Environmental
scientists

Secondary Users

General public, NGO untuk
environmental monitoring,
Mahasiswa untuk citizen science

Geographic Scope

Indonesia, khususnya daerah rural
& disaster-prone areas

Usia/Demografis	20-55 tahun, tech-literate, dengan access ke smartphone
Volume Pengguna	MVP: 100-500 pengguna; Scale-up: 5,000+ pengguna dalam 1 tahun

5. DESKRIPSI IDE

A. Konsep Solusi: Penjelasan Umum Ide

Water Presence Monitoring System adalah platform terintegrasi yang menggabungkan tiga sumber data untuk memberikan penilaian keberadaan air permukaan secara real-time:

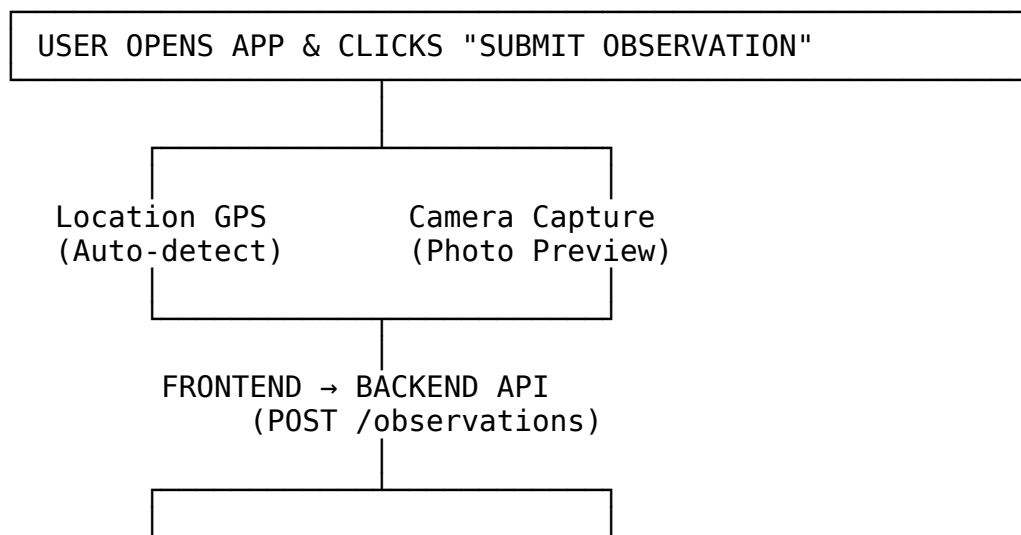
1. **Citizen Observations (Crowdsourced)**: Pengguna submit foto lokasi + GPS menggunakan smartphone
2. **AI Vision Analysis (Gemini 2.0 Flash)**: Analisis visual untuk mendeteksi kondisi tanah, kehadiran air, jenis vegetasi dengan confidence score 0-100%
3. **Satellite Validation (Sentinel-2 via Google Earth Engine)**: Perhitungan NDWI (Normalized Difference Water Index) untuk validasi keberadaan air permukaan
4. **Weather Data Integration (BMKG API)**: Korelasi dengan data cuaca untuk konteks kondisi air (curah hujan 24 jam, suhu, kelembaban)

Keluaran sistem: Confidence score 0-100% yang menunjukkan likelihood keberadaan air, disertai dengan breakdown sumber data, visualisasi di peta interaktif, dan risk indicators.

Waktu processing: 10-15 detik per observation (parallel processing via Bull queue system)

B. Rancangan Awal: User Flow dengan Penjelasan

User Flow Diagram:



PARALLEL PROCESSING (10-15 sec):

- Gemini Vision API (2-5s)
-> soil type, water presence, confidence
- Google Earth Engine (5-30s, cached)
-> NDWI calculation, cloud cover check
- BMKG Weather API (1-3s, optional)
-> temperature, humidity, rainfall

COMPARISON ENGINE

- Merge results from 3 sources
- Calculate final confidence score (0-100%)
- Generate verdict (High/Medium/Low/None)
- Detect anomalies (source disagreement, cloud cover)

DISPLAY TO USER

- Confidence gauge (0-100%)
- Water presence verdict
- Map with satellite overlay
- Risk indicators
- Historical comparison

Hubungan dengan Riset Pengguna:

- **Disaster Response Manager:** Mendapat alert + peta real-time dalam 15 detik untuk rapid assessment
- **Water Resource Manager:** Dapat track ketersediaan air per lokasi dengan confidence metrics untuk planning
- **Environmental Scientist:** Dapat participate dalam citizen science dengan transparent methodology

C. Fitur Utama Aplikasi

1. Observation Submission Form

- GPS auto-detection dengan opsi manual adjustment via drag pada map
- Native camera capture dengan real-time preview dan kompresi otomatis (target: max 2MB, 1920x1440)
- Opsional: time window selector untuk satellite data lookback
- Client-side image compression (ImageCompression.js) untuk mengurangi ukuran upload
- SHA256 hashing untuk deduplikasi foto

2. Results Dashboard (4-Panel Layout)

- Panel 1: Photo display dengan metadata lokasi dan timestamp
- Panel 2: Analysis summary (confidence gauge 0-100%, water presence verdict, risk indicators)
- Panel 3: Interactive map (location marker, satellite NDWI overlay, historical trend chart 30 hari)
- Panel 4: Detailed breakdown (Gemini analysis text, satellite data sources, weather context, comparison dengan historical baseline)

3. Explore/Map View

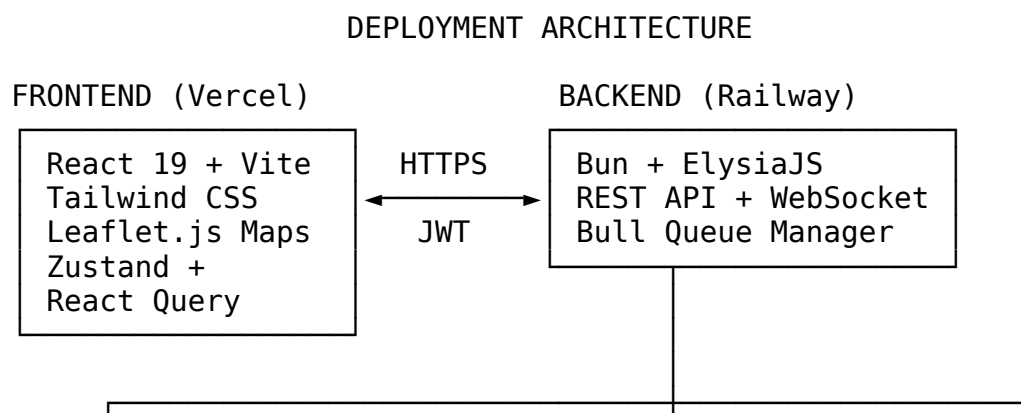
- Interactive map (Leaflet.js) dengan clustering observations per region
- Heatmap overlay (water presence probability per region menggunakan Leaflet.heat)
- Filters: date range, confidence threshold, water presence level
- Regional statistics & trends (time-series chart per province/district)
- Klik marker untuk melihat detail observasi

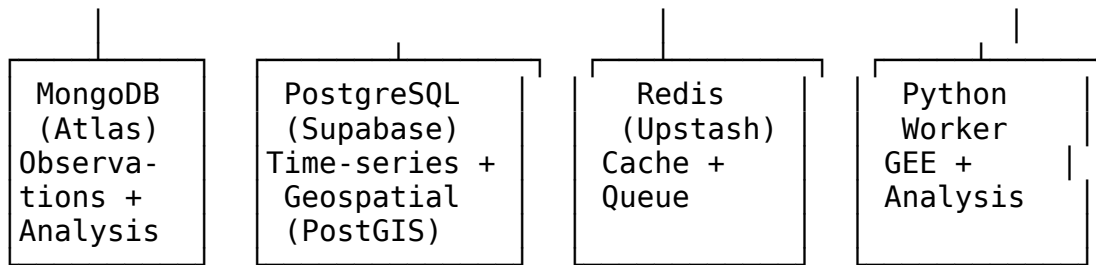
4. Methodology & About

- How the system works (transparent explanation with diagrams)
- Limitations & disclaimers (honest about 60-80% accuracy)
- Data sources attribution (Gemini, Sentinel-2, BMKG, OpenStreetMap)
- FAQ untuk pengguna umum

D. Arsitektur Sistem

Sistem dirancang dengan arsitektur microservices yang terpisah antara frontend, backend, dan worker processing:





Komponen Arsitektur:

- **Frontend:** React 19 + Vite dengan PWA capability (offline minimal). State management menggunakan Zustand untuk global state dan React Query untuk server state. Map integration menggunakan Leaflet.js dengan plugin heatmap dan GeoJSON overlay.
- **Backend API:** Bun runtime + ElysiaJS framework. REST API endpoints untuk authentication, observations, analysis results, dan explore/analytics. Middleware meliputi CORS, rate limiter (100 req/min per IP), JWT validation, centralized error handling.
- **Python Worker:** Microservice terpisah untuk Google Earth Engine queries (NDWI calculation) dan BMKG weather fetching. Berkomunikasi dengan backend via HTTP.
- **Queue System:** Bull (Redis-backed) untuk async job processing dengan 5 job types: photo analysis, satellite data, weather data, comparison engine, dan regional aggregation (hourly batch).
- **Caching Strategy:** Satellite data di-cache 7 hari, weather data 1 jam, regional aggregates 6 jam. Analysis results tidak pernah expire (historical record).

E. Rancangan Database

MongoDB (Observations)

Flexible schema untuk rapidly evolving data:

- **observations:** userId, lokasi (lat/lng), photoUrl, status (pending/processing/completed/error)
- **analysis_results:** gemini_result (soilType, waterPresence, confidence), satellite_data (ndwiValue, cloudCover, hasWaterFeature), weather_data, analysis (confidence_score, verdict, anomalies)

PostgreSQL + PostGIS

Time-series dan geospatial queries untuk analytics:

- time_series_observations: ndwi_value, confidence_score, water_presence (indexed by date)
- spatial_index: geom (PostGIS geometry), region (province/district), indexed untuk geographic queries

Sync Strategy: MongoDB sebagai source of truth, PostgreSQL sebagai denormalized copy dengan eventual consistency via change streams.

F. API Endpoints

Method	Endpoint	Deskripsi
POST	/api/v1/auth/register	Registrasi pengguna
POST	/api/v1/auth/login	Login pengguna
POST	/api/v1/observations	Submit observasi baru (multipart: foto + GPS)
GET	/api/v1/observations/:id	Detail observasi
GET	/api/v1/observations/:id/analysis	Hasil analysis (di-polling frontend tiap 2 detik)
GET	/api/v1/observations/:id/satellite-data	Data satelit mentah
GET	/api/v1/observations/:id/weather-data	Data cuaca
GET	/api/v1/map/heatmap	GeoJSON heatmap untuk region
GET	/api/v1/map/clusters	Clustered observations
GET	/api/v1/regions/:region/stats	Statistik regional
GET	/api/v1/regions/:region/trends	Time-series data regional
GET	/api/v1/health	System health check

G. Rancangan Teknis Singkat

Frontend	React 19 + Vite + Tailwind CSS + Leaflet.js + Zustand + React Query
Backend	Bun runtime + ElysiaJS framework
AI/Vision	Google Gemini 2.0 Flash (photo analysis)
Satellite	Google Earth Engine (Sentinel-2 NDWI)
Weather	BMKG API (primary) + OpenWeather (fallback)
Database	MongoDB Atlas (observations) + PostgreSQL Supabase (time-series)
Cache/Queue	Redis Upstash + Bull (async job processing)
File Storage	AWS S3 / DigitalOcean Spaces (image storage)
Monitoring	Sentry (error tracking) + Winston (logging)
Deployment	Vercel (frontend) + Railway (backend + Python worker)

6. TUJUAN

1. **Percepatan Deteksi:** Mengurangi waktu deteksi keberadaan air dari 3-5 hari (survey manual) menjadi 10-15 detik (sistem otomatis)
2. **Aksesibilitas:** Membangun tool yang accessible via smartphone saja, tanpa butuh expertise khusus atau peralatan mahal
3. **Akurasi:** Mencapai confidence score yang reliable (F1-score 0.65-0.80) melalui validasi multi-sumber (AI vision + satellite + weather)
4. **Skalabilitas:** Menghasilkan platform yang dapat scale ke ribuan pengguna dan observasi per hari dengan serverless architecture
5. **Transparansi:** Provide confidence metrics & methodology disclosure sehingga pengguna understand keterbatasan sistem
6. **Dampak Sosial:** Mendukung disaster response, agricultural planning, dan environmental monitoring di Indonesia

7. INOVASI

Keunikan Ide dibanding Solusi Existing:

1. **Multi-Source Validation Architecture**

- Existing tools: Gunakan single source (satellite saja, atau photo saja)
- Kami: Kombinasi 3 sumber (Gemini Vision + Sentinel-2 NDWI + Weather) dengan weighted confidence scoring

2. Real-Time Citizen Science Platform

- Existing: Citizen observations biasa dikumpul manual atau via form email (lambat)
- Kami: Smartphone app dengan instant processing & automated satellite validation dalam 10-15 detik

3. Confidence Scoring & Transparency

- Existing: Output biner (ada/tidak ada air)
- Kami: Confidence score 0-100% dengan breakdown per sumber data, coherence penalty, anomaly detection

4. Integrated AI + Satellite Processing

- Existing: Satellite processing biasa butuh expertise (GIS, remote sensing)
- Kami: Automated via Google Earth Engine API + Gemini Vision, accessible untuk non-expert

5. Cost-Effective Solution

- Existing: Professional survey: 5-10M per lokasi, Satellite imagery: 1-5M per order
- Kami: Smartphone camera (gratis) + APIs dengan free/cheap tier (Gemini: \$0.0008/image, GEE: free tier) → accessible untuk community-level

8. DAMPAK IDE/PROYEK

Dampak Langsung (Direct Impact):

1. Disaster Response (BNPB, disaster managers)

- Reduce assessment time dari hari menjadi menit
- Better spatial mapping untuk prioritas evakuasi
- Estimated: Bisa accelerate response 2-3 hari lebih cepat per event bencana

2. Agricultural Planning (Dinas Pertanian, farmers)

- Real-time indicator ketersediaan air untuk irrigation planning

- Reduce crop loss dari drought via early detection
- Estimated: +10-15% yield improvement di dry seasons

3. **Environmental Monitoring** (Researchers, NGOs)

- Baseline data collection untuk water resources assessment
- Crowdsourced validation untuk environmental projects
- Estimated: 5x lebih banyak data points dibanding traditional surveys

Dampak SDGs (Sustainable Development Goals):

- **SDG 6 (Clean Water & Sanitation):** Improve monitoring & management of water resources
- **SDG 13 (Climate Action):** Support disaster preparedness & response dengan data real-time
- **SDG 15 (Life on Land):** Environmental monitoring untuk conservation dan ecosystem management

Dampak Jangka Panjang:

- Foundation untuk integrated water resource management system di Indonesia
- Scalable model untuk citizen science platform di bidang environment yang bisa diadopsi negara lain
- Potential integration dengan government agencies (BNPB, PUPR, Kementan)
- Dataset crowdsourced yang dapat digunakan untuk penelitian dan kebijakan publik

9. ASPEK STEM

	Aplikasi ilmu Hydrology & Remote Sensing: NDWI (Normalized Difference Water Index) = $(\text{Green} - \text{NIR}) / (\text{Green} + \text{NIR})$ menggunakan Sentinel-2 band B3 (Green) dan B8 (NIR) untuk deteksi water bodies dari multispectral imagery. Threshold NDWI ≥ 0.3 mengindikasikan keberadaan air. Environmental science principles untuk data interpretation dan korelasi dengan data cuaca.
Science	
Technology	Full-stack modern development: React 19 + Vite (frontend), Bun/ElysiaJS (backend), Python microservice (satellite processing). Cloud infrastructure: Vercel, Railway, MongoDB Atlas, Supabase. APIs: Google Gemini Vision 2.0 Flash, Google Earth Engine, BMKG. Real-time data processing via Bull queue system dengan Redis.
Engineering	Software engineering: REST API design, database architecture (MongoDB + PostgreSQL dual-database), microservices pattern (Python worker terpisah untuk GEE processing). System integration: menghubungkan 3+ external APIs secara seamless. UI/UX design untuk accessibility di mobile dengan kompresi gambar client-side dan validasi form real-time.
Mathematics	Confidence score algorithm: weighted averaging dari multiple data sources. Formula: $\text{confidence} = (0.4 \text{ gemini_conf}) + (0.4 \text{ satellite_conf}) + (0.2 \text{ agreement_score})$. NDWI formula dengan normalized difference. Statistical analysis untuk accuracy metrics (precision, recall, F1-score). Time-series analysis untuk regional trends dan anomaly detection.

10. KONTRIBUSI AI

Peran AI dalam Sistem:

1. Gemini 2.0 Vision API (Photo Analysis)

- Input: Smartphone photo dari user (JPEG/PNG, max 5MB)
- Processing: Visual analysis untuk mendeteksi:
 - Soil type (clay, sandy, loam, rocky, peat)
 - Water presence (high/medium/low/none)
 - Surface conditions (wet, damp, dry, flooded, standing water)
 - Vegetation type & health
 - Anomalies & concerns
- Output: JSON structured response dengan confidence score 0-100
- Cost: USD 0.0008 per image (free tier: 50,000 images/month)
- Latency: 2-5 detik per call

2. Adaptive Processing Pipeline

- AI-driven decision logic untuk merging results dari 3 sources
- Automatic anomaly detection (e.g., high cloud cover flagged, source disagreement  40 points)
- Confidence score calculation dengan weighted formula dan coherence penalty
- Fallback mechanism jika salah satu source gagal (partial data tetap ditampilkan)

3. Pattern Recognition & Learning

- Historical trend detection per region (time-series analysis 30/90 hari)
- Seasonal pattern analysis untuk agricultural planning
- Potensi untuk future ML model training jika ada ground truth dataset yang cukup

Mengapa AI Critical:

- Manual photo analysis tidak scalable untuk ribuan observasi
- Gemini Vision memberikan analisis instant, standardized, dan reproducible
- Automated confidence scoring memastikan transparansi dan konsistensi
- AI-enabled anomaly detection memberikan early warning untuk kondisi tidak biasa

Detail Prompt Engineering untuk Gemini:

System Prompt:

"You are an environmental analyst specializing in water and soil assessment. Analyze the provided image and respond **ONLY** in valid JSON format with no markdown or preamble."

Response JSON:

```
{
  "soilType": "loamy clay",
  "waterPresence": "medium",
  "confidence": 72,
  "surface": "damp soil with standing puddles",
  "vegetation": "grass and reeds, healthy",
  "anomalies": ["muddy discharge visible, possibly from upstream"],
  "notes": "High water table indicated by vegetation and conditions"
}
```

11. SUSTAINABILITY

Strategi Keberlanjutan (Sustainability Strategy):

1. Business Model

- **Freemium:** Observations & basic results gratis untuk public users
- **Government Partnerships:** Premium subscription untuk BNPB, Dinas Pertanian, PUPR (dengan dedicated support, advanced analytics, API access)
- **NGO Programs:** Discounted rates untuk research & conservation projects
- **Estimated revenue:** IDR 500M - 1B per tahun di year 2-3

2. Technical Sustainability

- Serverless architecture (Vercel, Railway) auto-scaling, minimal ops overhead
- Open-source components (React, Leaflet, Tailwind) dengan strong community support
- API-first design memudahkan integrasi dengan third-party tools dan government systems

- Regular updates based on user feedback & technology trends
- Python worker dapat di-extend untuk additional data sources di masa depan

3. **Community Engagement**

- Citizen science program: reward active contributors (badges, leaderboards, contribution recognition)
- Educational outreach: training workshops untuk disaster managers, farmers, dan mahasiswa
- Partnership dengan universitas untuk research validation dan dataset development
- Transparent data sharing (CC-BY license) untuk credibility dan academic use

4. **Impact Measurement**

- Monthly KPIs: observation count, unique users, accuracy metrics, response time
- Quarterly user satisfaction surveys untuk continuous improvement
- Annual impact report: lives helped, disaster response time reduced, area monitored, publications

5. **Data Governance**

- Open data: All observations available via public API dengan CC-BY license
- Privacy-first: No personal data collected beyond email (optional) untuk authentication
- Data retention: 5 years on MongoDB, archival to cold storage (S3 Glacier) setelahnya
- Foto dihapus setelah 30 hari untuk menghemat biaya storage

INFORMASI TAMBAHAN

A. Timeline Pengembangan (Development Roadmap)

Week 1: Foundation & Setup (25-30 jam)

- Setup environment: GitHub repo, monorepo structure (frontend/ + backend/), Docker Compose (MongoDB, PostgreSQL, Redis)
- Frontend: React + Vite project, Tailwind CSS, Leaflet, routing, layout components, halaman Home
- Backend: ElysiaJS server skeleton, database models (Mongoose), REST routes (CRUD observations), error handling middleware

- Testing: Frontend → Backend integration, mobile responsive, Postman/curl verification

Week 2: AI & Satellite Integration (30-35 jam)

- Gemini Vision API: geminiService.ts wrapper, analyzePhoto job processor, error handling + retries
- Google Earth Engine: Service account setup, Python microservice, NDWI calculation function, Redis caching (7-day TTL)
- Data Flow Integration: 3 parallel jobs (Gemini + GEE + BMKG), job completion check, comparison trigger
- Tantangan: Gemini response format (force JSON), GEE learning curve, cloud cover filtering

Week 3: Analysis Engine & Results Page (30-35 jam)

- Comparison Engine: calculateConfidence() function, weighted scoring algorithm, verdict generator, anomaly detection
- Results Page UI: 4-panel layout (photo, analysis summary, map, detailed breakdown), polling dengan React Query
- Explore/Map View: Interactive map (Leaflet), heatmap overlay, filters (date range, confidence threshold)
- Backend Analytics: /map/heatmap endpoint (GeoJSON), /regions/:region/stats, regional aggregation

Week 4: Testing, Polish & Launch Prep (30-35 jam)

- Validation Testing: 5-10 real observations di berbagai environment (urban, agricultural, water body, dry area, wet area)
- Bug Fixes: Critical bugs, performance optimization, image compression, caching, database indices
- Documentation: README, API docs, methodology, limitations, setup guide, screenshots
- Deployment: Vercel (frontend), Railway (backend), production testing, rollback plan

Week	Total Hours	Key Deliverable
1	25-30	Functional form submit → DB save → return ID
2	30-35	End-to-end: submit → Gemini + GEE → stored

Week	Total Hours	Key Deliverable
3	30-35	Complete results page + explore map + analysis engine
4	30-35	Production-ready MVP, validated, deployed & tested
Total	115-135	~4 weeks full-time equivalent

B. Tech Stack Summary

Frontend	React 19 + Vite + Tailwind CSS + Leaflet.js + Zustand + React Query
Backend	Bun + ElysiaJS dengan middleware CORS, rate limiter, JWT
Python Worker	FastAPI microservice untuk GEE dan BMKG processing
Database	MongoDB Atlas (observations & analysis) + PostgreSQL Supabase (time-series + PostGIS) + Redis Upstash (cache & queue)
AI/APIs	Gemini 2.0 Flash, Google Earth Engine, BMKG API (fallback: OpenWeather)
Queue	Bull (Redis-backed, 5 job types, retry 2x, exponential backoff, dead-letter queue)
Image Storage	AWS S3 / DigitalOcean Spaces dengan hashed filename
Deployment	Vercel (frontend) + Railway (backend + Python worker)
Monitoring	Sentry (error tracking) + Winston (structured logging) + Prometheus (/metrics endpoint)

C. Data Processing Pipeline & Confidence Scoring

Request Lifecycle:

1. User submits observation (location + photo) via frontend
2. Backend validates input (location range, photo size < 5MB, file type JPEG/PNG, rate limit 10 observasi/hari)

3. Observation saved ke MongoDB dengan status “pending”
4. 3 parallel jobs di-queue via Bull:
 - **Job 1 - Photo Analysis** (2-5 detik): Encode foto ke base64, call Gemini Vision API, parse JSON response, simpan ke analysis_results.gemini_result
 - **Job 2 - Satellite Data** (5-30 detik): Query GEE untuk NDWI di lokasi, filter cloud cover $\leq$ 20%, cache 7 hari di Redis
 - **Job 3 - Weather Data** (1-3 detik): Fetch BMKG (timeout 5 detik), fallback ke OpenWeather jika gagal, cache 1 jam
5. Comparison Engine berjalan setelah semua job selesai:
 - Merge results, hitung confidence score, generate verdict
 - Deteksi anomalies: source disagreement $\leq$ 40 poin, cloud cover $\leq$ 50%, vegetation-soil mismatch
 - Update status observasi menjadi “completed”
6. Frontend polling (tiap 2 detik) via React Query, menampilkan results dalam 4-panel layout

Confidence Score Formula:

```
confidence = (0.40 x gemini_conf)
            + (0.40 x satellite_conf)
            + (0.20 x agreement_score)
```

dimana:

```
gemini_conf = confidence score dari Gemini Vision (0-100)
```

```
satellite_conf = {
  80 jika NDWI > 0.5 (definitely water)
  60 jika NDWI 0.3-0.5 (likely water)
  40 jika NDWI 0-0.3 (maybe water)
  20 jika NDWI < 0 (probably not water)
  0 jika tidak ada satellite data
}
```

```
agreement_score = {
  100 jika kedua source > 60 (both high)
  100 jika kedua source < 40 (both low)
  50 jika mixed (satu high, satu low)
  0 jika hanya satu source available
}
```

Water Presence Verdict:

Confidence	Verdict
≥ 75	HIGH (warning: surface water likely present)
50 - 74	MEDIUM (caution: possible water, verify manually)
25 - 49	LOW (info: limited water detected, continue monitoring)
 25	NONE (clear: no water detected)

D. Error Handling & Graceful Degradation

Sistem dirancang untuk tetap berfungsi meskipun ada komponen yang gagal:

Skenario	Response Sistem
Gemini Vision gagal	Status “completed with partial data”, tampilkan satellite + weather, note: “Photo analysis unavailable”
Satellite data unavailable	Status “completed with partial data”, tampilkan Gemini + weather, note: “No recent satellite imagery (cloudy or unavailable)”
Keduanya gagal	Status “completed with warnings”, tampilkan weather only, rekomendasi manual survey
BMKG down	Skip weather panel, mark “Weather data unavailable”, analysis tetap dilanjutkan
Redis down	Queue jobs bypass Redis, process synchronously (slower but functional)
Photo terlalu besar	Client-side compression (ImageCompression.js), max 2MB setelah kompresi

E. Accuracy & Validation Strategy

- **Expected Accuracy:** F1-score 0.65-0.80 untuk MVP (reasonable untuk citizen science platform)
- **Validation Method:** Compare system output vs ground truth di 5-10 lokasi berbeda
- **Test Environments:**
 - Urban area (buildings, asphalt, parking lot)

- Agricultural field (soil, crops, irrigation)
- Water body (lake, river, pond)
- Dry area (desert, concrete, dry field)
- Wet area (swamp, flooded field, wetland)
- **Accuracy Metrics:** Precision = $TP/(TP+FP)$, Recall = $TP/(TP+FN)$, $F1 = 2 \times (P \times R)/(P + R)$
- **Target:** $F1 \geq 0.70$ dengan error rate $\leq 5\%$ pada load test 100 simultaneous observations
- **Limitations:** Cloud cover (30-50% images unusable), satellite data lag (3-5 days), phone photo quality dependency, shadow confusion di area pegunungan
- **Transparency:** Confidence score breakdown ditampilkan ke user, limitations clearly stated di halaman Methodology

F. Testing & Deployment Checklist

Pre-Launch Checklist:

- Frontend: all pages tested on mobile (iOS/Android), responsive design verified
- Backend: all endpoints tested with proper error codes, rate limits configured
- Database: indices created, backup schedule configured (daily MongoDB dump + PostgreSQL dump)
- APIs: rate limits set (Gemini 50k/month, GEE quotas managed), caching configured
- Security: JWT refresh mechanism working, CORS configured, file upload validation
- Load test: simulate 100 simultaneous observations, monitor queue length & response time
- Rollback plan: previous Docker image tagged, database snapshots available

Launch Day Checklist:

- Deploy frontend ke Vercel (auto CI/CD dari GitHub)
- Deploy backend ke Railway (Docker container)
- Monitor Sentry untuk errors real-time
- Test end-to-end dengan sample observation di production
- Monitor queue length via Bull dashboard
- Backup pre-recorded demo jika WiFi gagal

G. Limitations & Honest Disclaimers

Apa yang sistem ini TIDAK bisa lakukan:

- **Deteksi groundwater** – Only surface water visible to satellites dan foto
- **Prediksi masa depan** – Only analyzes current moment, no forecasting capability
- **Menggantikan professional surveys** – Accuracy 60-80%, not 99%
- **Bekerja tanpa internet** – Requires cloud APIs (Gemini, GEE, BMKG)
- **Membedakan jenis air** – Saltwater vs freshwater tidak dibedakan
- **Mengukur kedalaman air** – Only presence/absence detection
- **Real-time sempurna** – Satellite data minimal 3-5 hari, tidak real-time absolute

Faktor yang mempengaruhi akurasi:

- Cloud cover: 30-50% of satellite images tidak dapat digunakan
- Time of day: Shadows mempengaruhi Gemini Vision analysis
- Image quality: Blurry photos reduce accuracy signifikan
- Terrain complexity: Mountains hide small water bodies
- Vegetation density: Dense plants obscure ground truth

H. Future Enhancements (Post-MVP)

Phase 2:

- User accounts with profile + observation history + social features
- API publik untuk third-party integrations
- Mobile app native (React Native / Flutter)
- Real-time alert system untuk anomaly detection

Phase 3:

- IoT sensor integration (optional physical sensors untuk ground truth)
- ML model trained on ground truth data (custom confidence model)
- Government agency integration (BNPB, PUPR, Kementan data sharing)
- Offline mode dengan local satellite caching

Phase 4:

- Forecasting model (predict water presence 7 days ahead)

- Thermal imaging analysis (jika pengguna punya thermal cameras)
- AR visualization (lihat water bodies melalui phone camera)
- Integration with drone imagery untuk higher resolution

I. Competitive Analysis

Feature/ Kriteria	Kami	Satellite Only	Photo Only	Manual Survey
Real-time	✓	×(days)	✓	×(days)
Accessible	✓(smartphone)	×(costly)	✓	×(expert needed)
Scalable	✓	×	✓	×
Validated	✓(multi-source)	✓	△(single)	✓
Cost/ observation	Rp 0	1-5M	0	5-10M
Accuracy	60-80%	70-85%	40-60%	95%+

J. Risk Mitigation

Risk	Like-lihood	Impact	Mitigation
Gemini quota exceeded	Medium	High	Cache results, batch processing, upgrade tier jika perlu
GEE queries slow	Medium	Medium	Cache 7 hari, set 30-sec timeout, fallback ke cached data
BMKG API down	High	Low	Skip gracefully, continue without weather, fallback ke OpenWeather
Satellite no data	Medium	Medium	Show “no recent satellite data”, use historical if available
Database corruption	Low	High	Daily backups (MongoDB dump)

Risk	Like-lihood	Impact	Mitigation
			+ PostgreSQL dump), test recovery monthly

K. References & Inspiration

- NDWI methodology: McFeeters, S. K. (1996). The use of Normalized Difference Water Index for discrimination of standing water from soil and vegetation
- Citizen science frameworks: Bonney, R., et al. (2009). Citizen Science: A Tool for Integrating Studies of Ecosystems
- Self-Determination Theory: Ryan, R. M., & Deci, E. L. (2000). Intrinsic and Extrinsic Motivations
- Google Earth Engine: Gorelick et al. (2017). Google Earth Engine: Planetary-scale geospatial analysis for everyone
- Sentinel-2 Band Configuration: ESA Copernicus Sentinel-2 documentation
- BMKG API: Badan Meteorologi, Klimatologi, dan Geofisika Indonesia - data.bmkg.go.id
- OpenStreetMap: Tile layer for map visualization (openstreetmap.org)

Proposal ini disusun dengan fokus pada aksesibilitas, transparansi, dan dampak sosial untuk mendukung tujuan pembangunan berkelanjutan di Indonesia. Sistem menggabungkan citizen science, AI vision, dan satellite remote sensing dalam satu platform terintegrasi yang accessible melalui smartphone.

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Mei 2026